



9 ways MOQ changes Media Delivery

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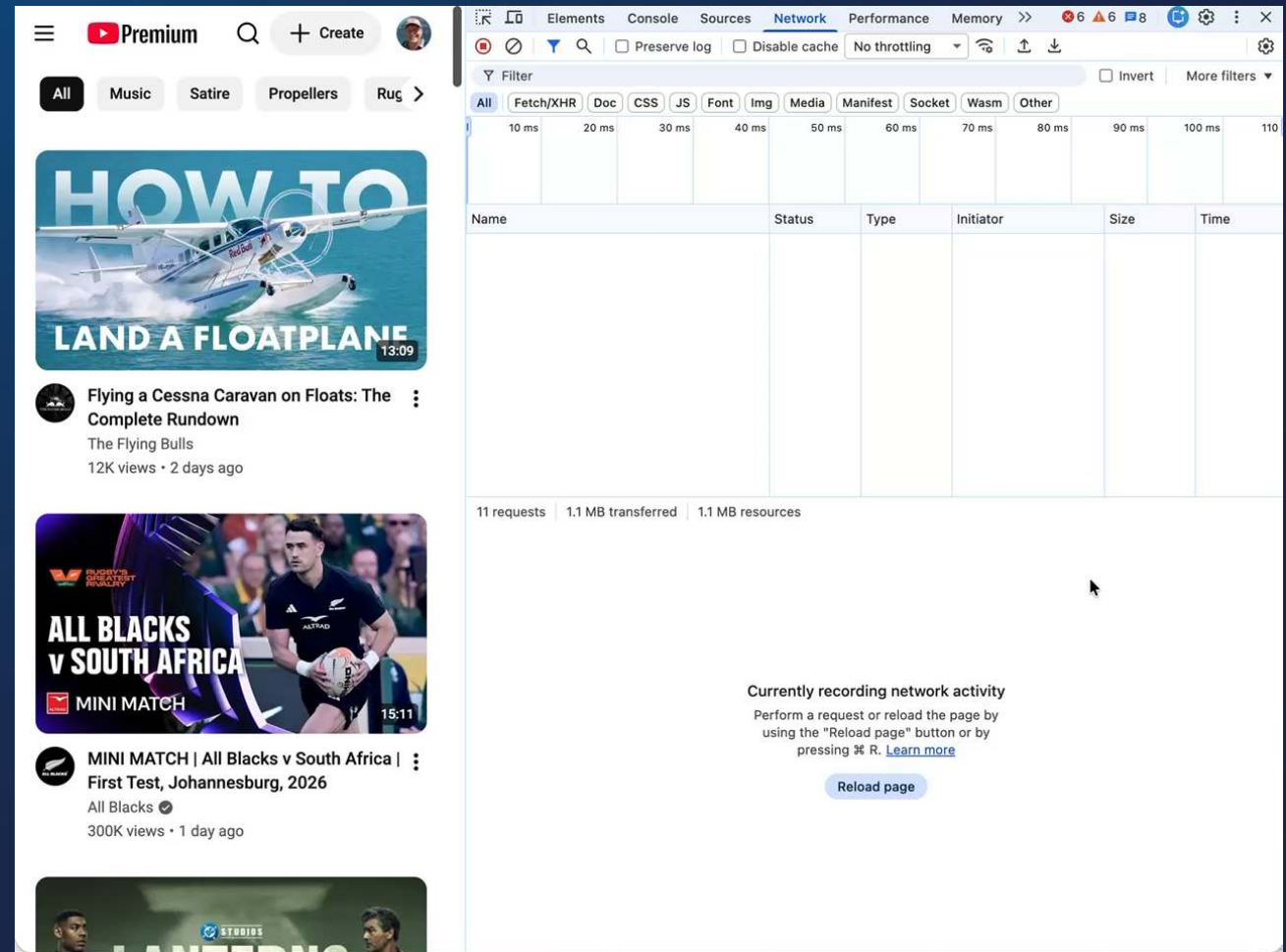
Sept 2026

1. MOQ isn't only for media

The core MOQ Transport spec is intentionally **payload agnostic**.

MOQT can move media well, but it can also *move all the other data associated with modern media playback*:

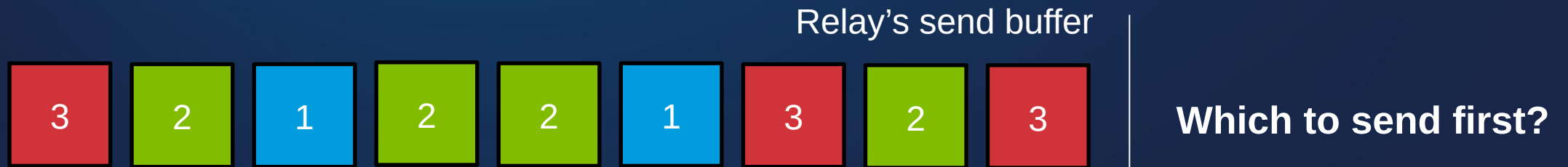
- Manifests
- Sports scores
- User interactivity
- Token refresh instructions
- QoE Data
- Log files
- Advertising decisioning



2. Priorities between flows in the same connections

MOQT allows **relative priorities** to exist between streams and datagrams, and to be defined separately both publishers and subscribers.

Business impact: prioritize audio over video, or base layer SVC over enhancement layers.



Publishers ranks:

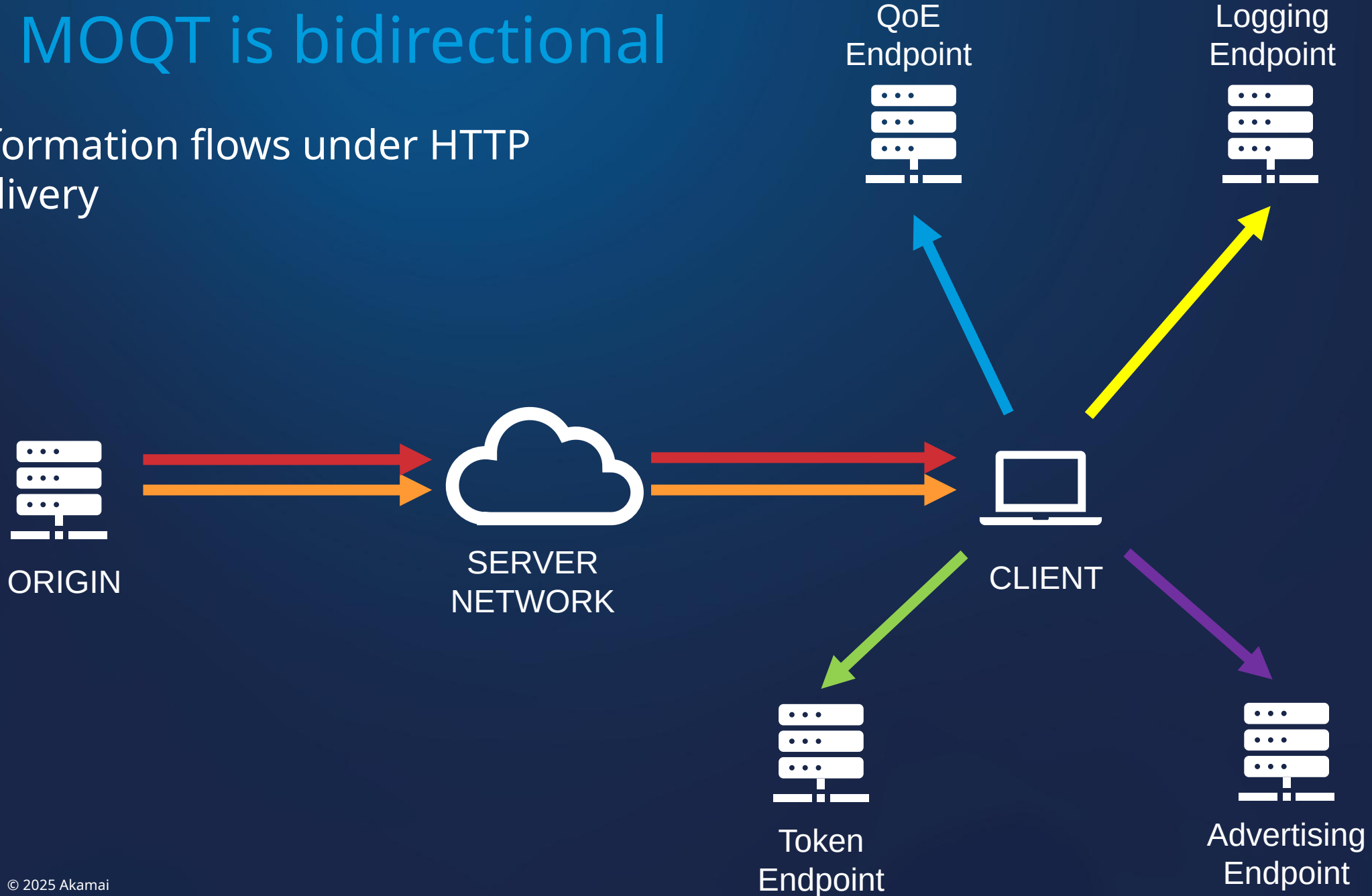
Blue - highest priority
Green - medium priority
Red - low priority

Subscriber ranks:

Green - highest priority
Blue - medium priority
Red - low priority

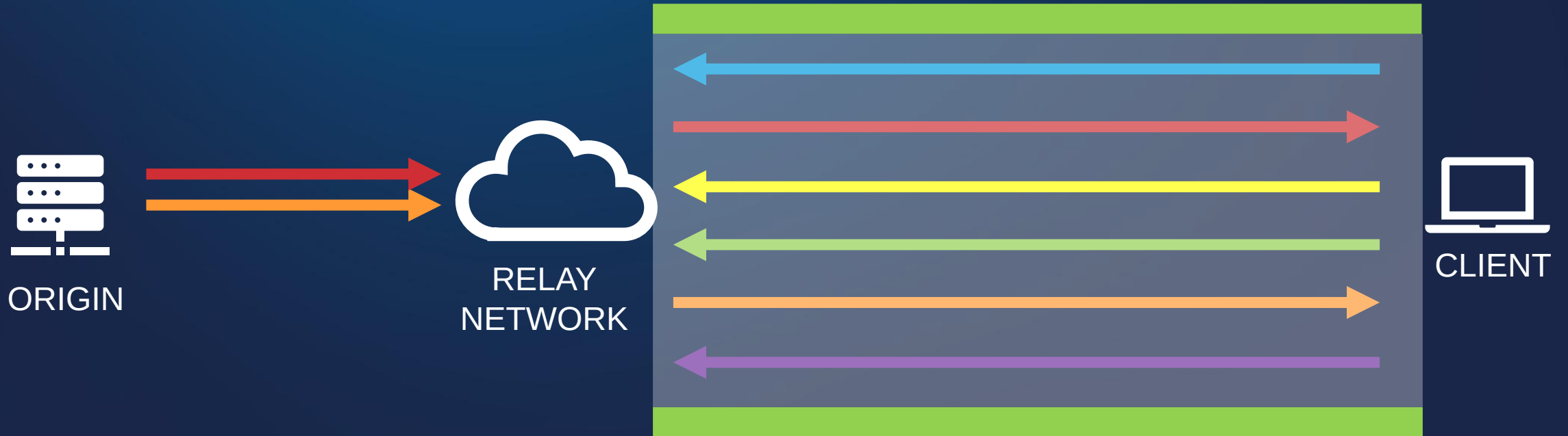
5. MOQT is bidirectional

Information flows under HTTP delivery



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Information flows under MOQT



Single QUIC TLS 1.3 connection

3. Network filtering

With MOQT, the distribution network can **intelligently filter** the incoming data using **Object Filters**

Use cases:

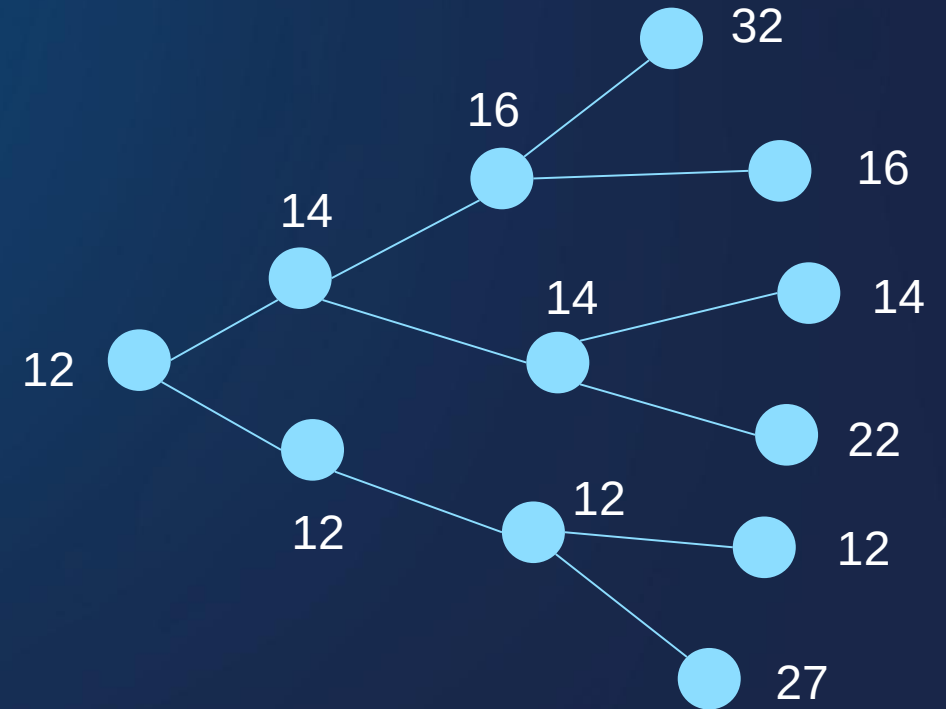
Meetings 100 people online, see/hear last 4 active speakers

Security Cams 100 cams, monitor top 6 with most activity

Viral Videos 100 live video feeds, see top 4 most popular

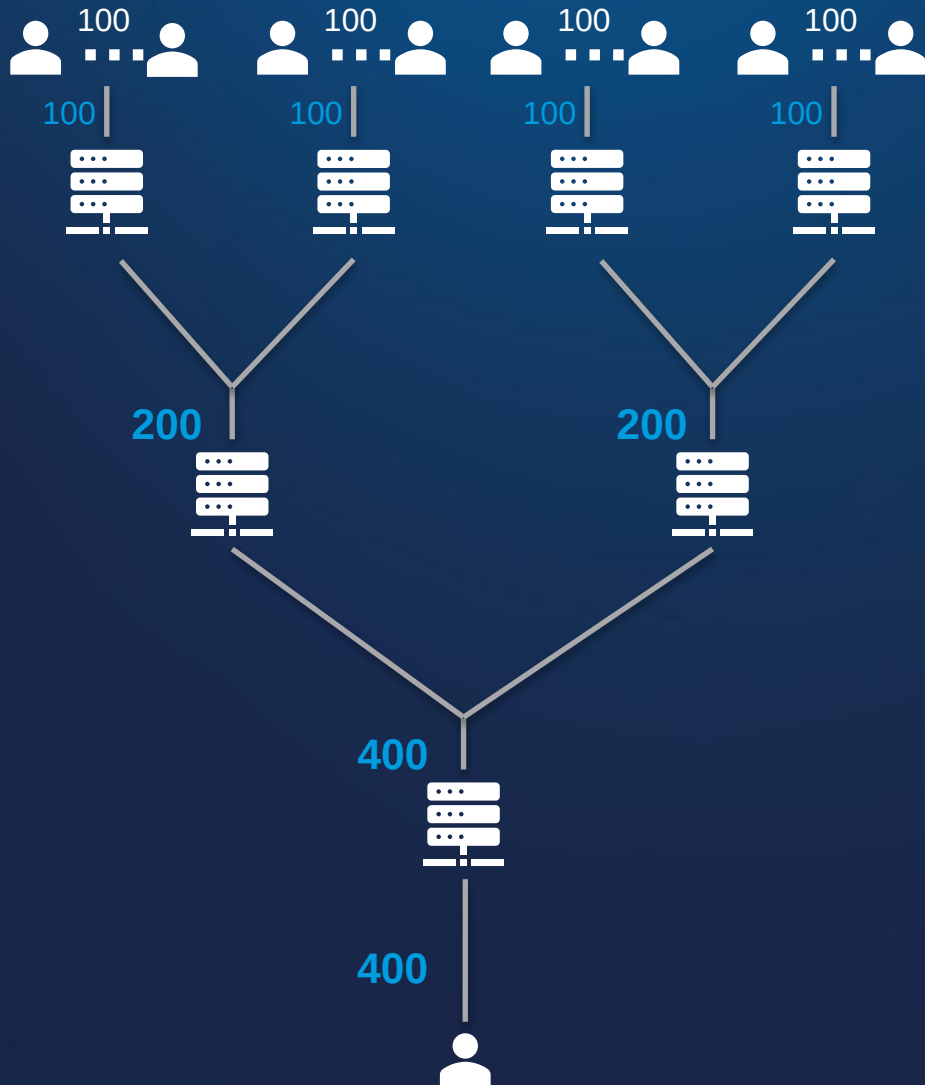
Sensors, Telemetry 100 temperature sensors, get top 4 hottest

Chat Tracks show me the last 25 comments from 10,000 chat clients.

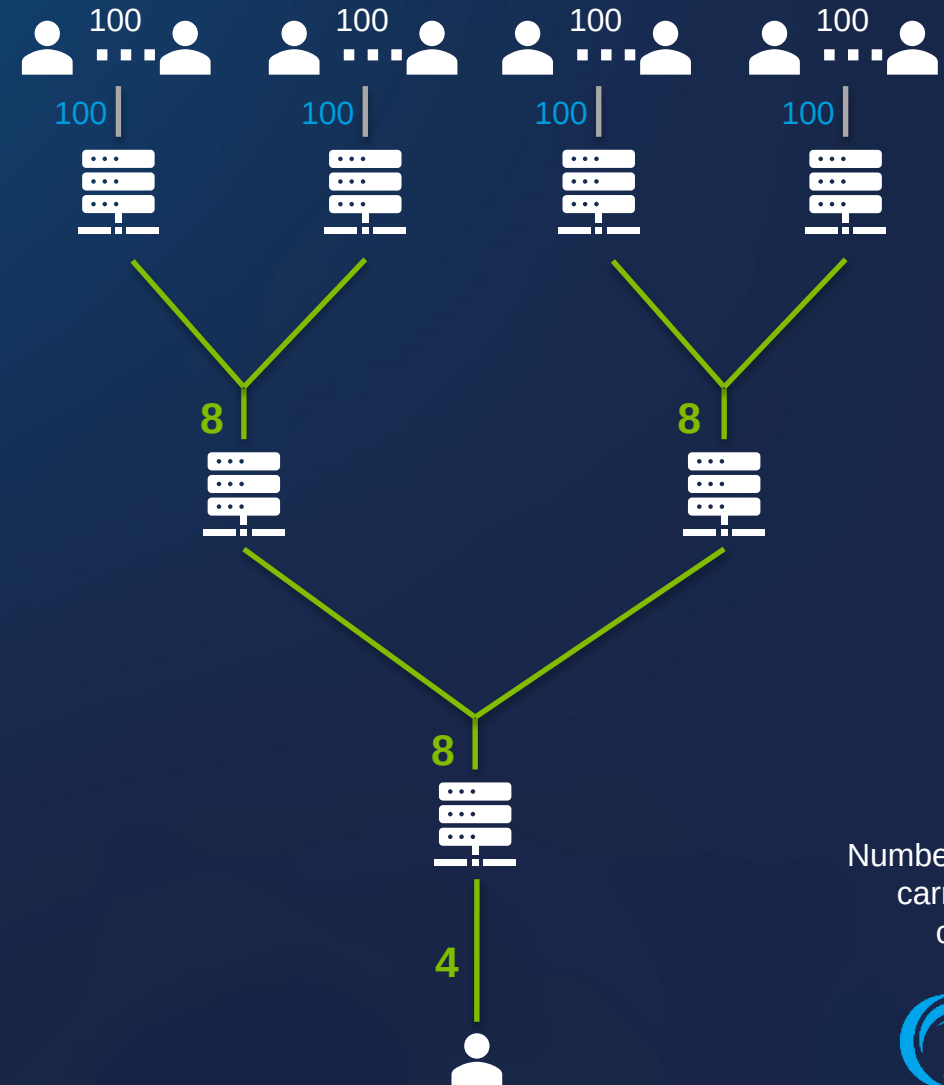


CDN midgress traffic for fan-in with & without top-N filtering

Without top-N in place at each tier

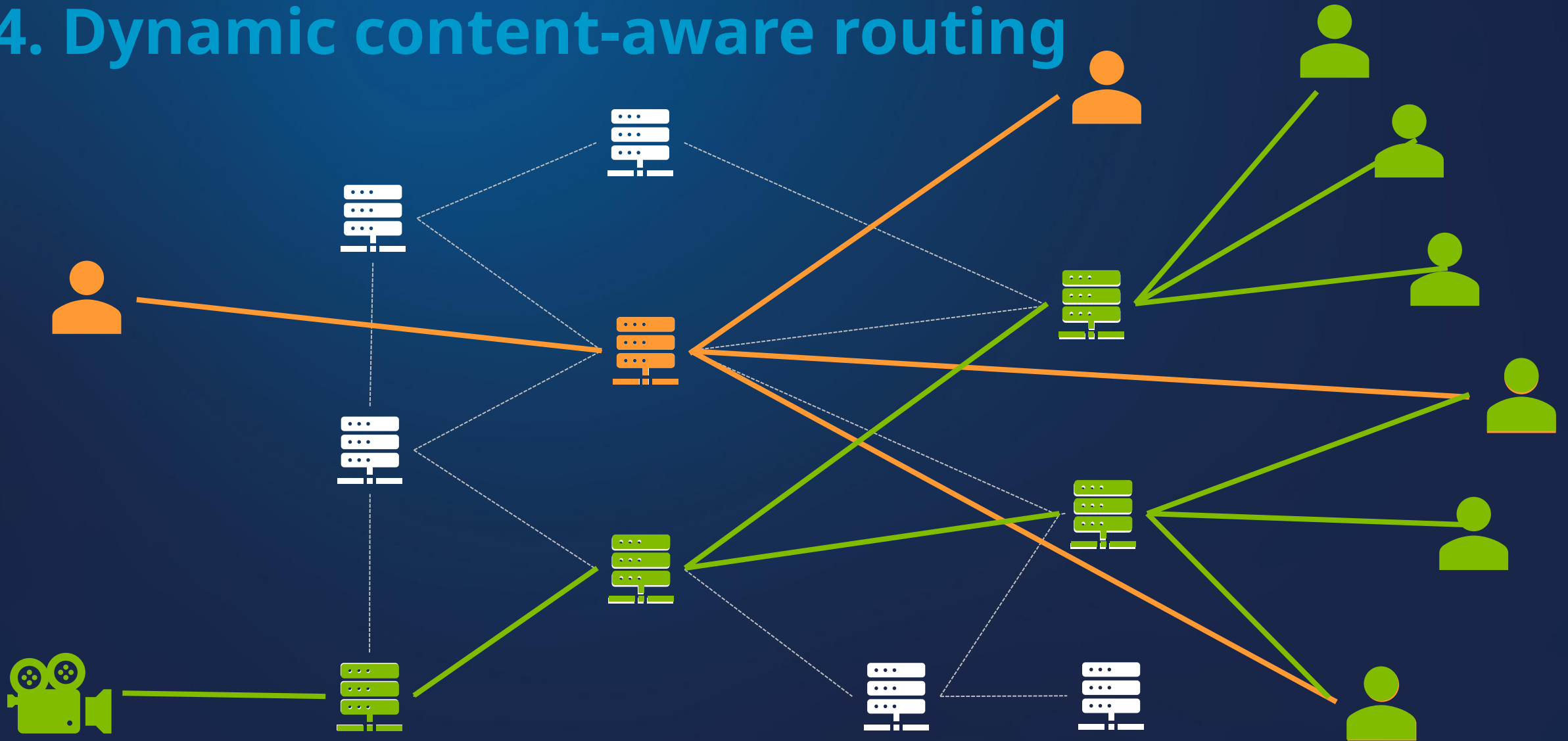


With top-4 in place at each tier



Number of tracks
carried by the
connection

4. Dynamic content-aware routing



Shortest path for real-time media

6. Edge prewarming via SUBSCRIBE forward parameter

A Subscription can request a publisher to not forward Objects for a given track by setting the Forward field to 0.

Use SUBSCRIBE_UPDATE to change the forwarding state.

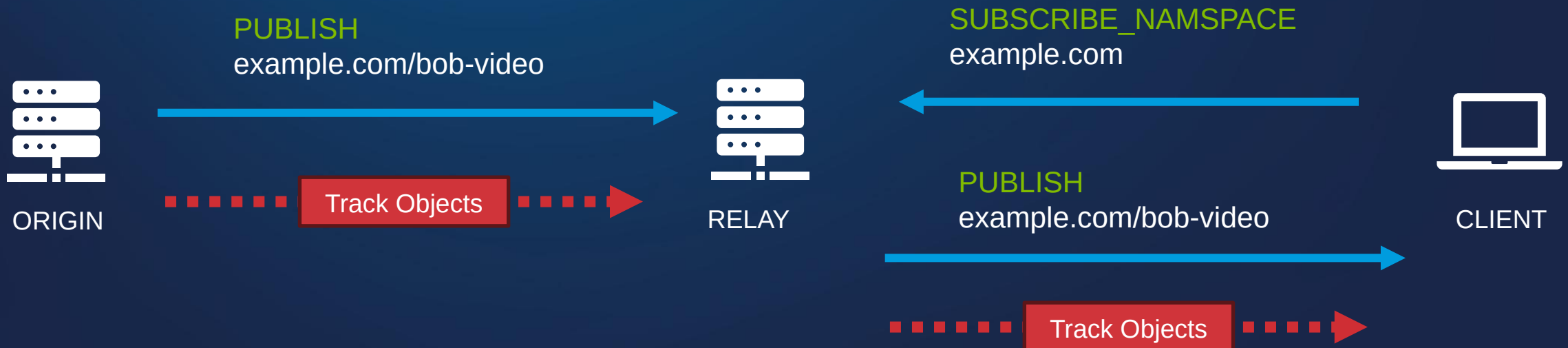


Useful for quick ABR switching as alternate tracks are pre-warmed at the edge.

Also a DDOS vector so relays may refuse the request.

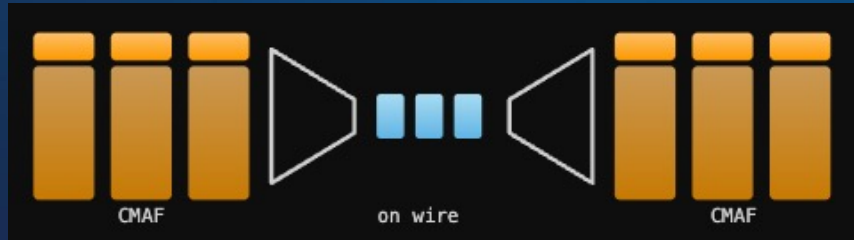
7. SUBSCRIBE to things you don't yet know

Allows receipt of all tracks produced under a namespace, without having to know their names. Useful if the track names are generated on the-fly, for example, people joining a web conference



8. Better CMAF - LOCMAF (low overhead CMAF for MOQ)

Hugo Björs (MSc thesis) · Torbjörn Einarsson · Eyevinn · 2026



- A new, compact wire format for streaming CMAF over MOQ Transport
- A typical CMAF chunk header for one sample is around ~104 bytes.
- LOCMAF observes that consecutive moof boxes are almost identical and delta-encodes them.



<https://locmaf.dev/>



track	sample	CMAF	LOCMAF	saved
audio_128kbps_aac	128.0 kbps	171.5 kbps	131.9 kbps	23.1 %
video_400kbps_avc	373.2 kbps	396.4 kbps	376.5 kbps	5.0 %

9. Shared software development OpenMOQ



- A software consortium founded to develop high performance, open-source MOQ software.
- Founded January 2026 – now have 20 members
- Get the software at <https://github.com/openmoq>



CDN77



ORACLE



Synamedia



YouTube



Fishjam



MAINSTREAMING®



nano
COSMOS



VINDRAL



WOWZA



UNIVERSITÄT
KLAGENFURT





Thank you for your time.